Final Exam

Oscar I. Hernandez

Dec 7, 2020

1. Consider the IBVP:

$$u_{tt} - u_{xx} = 0,\ u(0,t) = 0,\ u(x,0) = 0, u_t(x,0) = f(x), \tag{1}$$

where $x > 0$, $t > 0$, $f(x) = 1$ for $0 \leq x \leq 1$ and $f(x) = 0$ for $x > 1$.

Proof (Hernandez). The solution is given by d'Alembert's formula below:

$$\begin{aligned} u(x,t) &= \frac{1}{2}\left[0+0\right] + \frac{1}{2\cdot 1}\int_{x-t}^{x+t} f(\xi)\ d\xi \\ &= 0 + \frac{1}{2}\int_{\min(x-t,0)}^{\min(x+t,1)} f(\xi)\ d\xi. \end{aligned}$$

□

2.

$$u_t - u_{xx} = \frac{1}{x^2+1},\ \ -\infty < x < \infty, t > 0, u(x,0) = 0. \tag{2}$$

Proof (Hernandez). According to Duhamel's principle, the solution is given below:

$$u(x,t) = \int_0^t \int_{-\infty}^{\infty} \frac{G(x-y, t-\tau)}{y^2+1}\ dyd\tau.$$

□

3.

$$u_{tt} - u_{xx} = x,\ u(x,0) = 0,\ u_t(x,0) = 1, u_x(0,t) = u_x(\pi,t) = 0, \tag{3}$$

where $0 < x < \pi$, $t > 0$.

Proof (Hernandez). The solution is given by a superposition $u = v + w$ solving:

$$\begin{aligned} v_{tt} - v_{xx} = 0,\ v(x,0) = 0,\ v_t(x,0) = 1, v_x(0,t) = v_x(\pi,t) = 0, \\ w_{tt} - w_{xx} = x,\ w(x,0) = 0,\ w_t(x,0) = 0, w_x(0,t) = w_x(\pi,t) = 0, \end{aligned}$$

Clearly, $v(x,t) = t$. For w, we obtain the following eigenfunctions:

$$\begin{aligned} -\varphi_n''(x) &= \lambda^2\varphi_n(x), & \varphi_(0) &= \varphi(\pi) = 0, \\ \varphi_n(x) &= \sin(\lambda x), & \lambda &= \frac{n\pi\cdot 1}{\pi} = n, \end{aligned}$$

which leads to the following solution:

$$w(x,t) = \sum_{n=1}^{\infty}\left(\frac{1}{n}\int_0^t f_n(s)\cos(ns)\ ds\right)\sin(nt)\sin(nx),$$

where $f_n(s)$ is the Fourier coefficient of x as shown below:

$$x = \sum_{n=1}^{\infty} f_n(t)\sin(nx).$$

□

4.

$$u_{tt}+u_{xxxx}=0, u(0,t)=u_{xx}(0,t)=0, u_x(1,t)=u_{xxx}(1,t)=0, u(x,0)=1, u_t(x,0)=\sin(3\pi x/2), \tag{4}$$

where $0<x<1$, $t>0$.

Proof (Hernandez). We obtain the following solution:

$$u(x,t)=\sum_{n=0}^{\infty}\frac{1}{n}\int_{-\infty}^{\infty}\frac{\sin(3\pi w/2)}{w^2}\sin(3\pi wt/2)e^{wtx}\ dw.$$

□

5.

$$u_{tt}=u_{xx}-2u,\ u_x(0,t)=0, u(1,t)=0, u(x,0)=1, u_t(x,0)=0, \tag{5}$$

where $0<x<1$, $t>0$.

Proof (Hernandez). The solution is given by the superposition $u=v+w$ solving:

$$\begin{aligned} v_{tt}-v_{xx}&=-2v,\ v_x(0,t)=0, v(1,t)=0, v(x,0)=0, v_t(x,0)=0,\\ w_{tt}-w_{xx}&=0,\ w_x(0,t)=0, w(1,t)=0, w(x,0)=1, w_t(x,0)=0, \end{aligned}$$

where $v=0$ everywhere and w is given by d'Alembert's formula below:

$$w(x,0)=\frac{1}{2}\left[1+1\right]+\frac{1}{2\cdot 1}\int_{x-ct}^{x+ct}0\ d\xi=1.$$

□

6.

$$\frac{1}{2x}u_x+u_t-u=0,\ \ -\infty<x<\infty, t>0, u(x,0)=x^2 \tag{6}$$

Proof (Hernandez). We rewrite this equation as follows:

$$u_t+c(x,t)u_x=f(x,t,u),$$

where $c(x,t)=c(x)=\frac{1}{2x}$ and $f(x,t,u)=u$, and transform to characteristic coordinates $U(\xi,\tau)$ where $\tau=t$ and $\xi=\phi(x,t)=C=x^2-t$ solves the following ODE:

$$x^2=\int 2x\cdot\frac{dx}{dt}\ dt=\int 2x\cdot\frac{1}{2x}\ dt=t+C.$$

Then, we obtain the following solution:

$$\begin{aligned} [-U_\xi+U_\tau]+\frac{1}{2x}\left[2x\cdot U_\xi\right]=U_\tau=U &\qquad \text{by substitution and chain rule}\\ U(\xi,\tau)=e^\tau\xi &\qquad \text{by anti-differentiation}\\ u(x,t)=e^t(x^2-t) &\qquad \text{re-coordinatization.} \end{aligned}$$

□

7.

$$(x+t)u_x + tu_t + u = 0, \ \ -\infty < x < \infty, \ t > 1, \ u(x,1) = x. \tag{7}$$

Proof (Hernandez). We rewrite this equation as follows:

$$u_t + c(x,t)u_x = f(x,t,u),$$

where $c(x,t) = \frac{x+t}{t}$ and $f(x,t,u) = -\frac{u}{t}$, and transform to characteristic coordinates $U(\xi,\tau)$ where $\tau = t$ and $\xi = \phi(x,t) = C = x/t - \ln(t)$ solves the following ODE:

$$\begin{aligned} \frac{dx}{dt} &= c(x,t) = \frac{x}{t} + \frac{t}{t} \\ \frac{1}{t}\frac{dx}{dt} - \frac{x}{t^2} = \left[\frac{dx}{dt} - \frac{x}{t}\right] e^{\int -\frac{1}{t}} &= \left[\frac{1}{1}\right] e^{\int -\frac{1}{t}} = 1 \cdot \frac{1}{t} \\ x/t &= \int \frac{d\left(\frac{x}{t}\right)}{dt} = \int \frac{1}{t}\, dt = \ln(t) + C \end{aligned}$$

Then, we obtain the following solution:

$$\begin{aligned} \left[U_\tau - \frac{x+t}{t^2}U_\xi\right] + \frac{x+t}{t}\left[\frac{1}{t}U_\xi\right] = U_\tau = -\frac{1}{t}U & \qquad \text{by substitution and chain rule} \\ U(\xi,\tau) = \frac{\xi}{\tau} & \qquad \text{by anti-differentiation} \\ u(x,t) = \frac{x/t - \ln(t)}{t} = \frac{x - \ln(t)t}{t^2} & \qquad \text{by re-coordinatization.} \end{aligned}$$

□

8. Consider the following IBVP in the unit 3-d sphere:

$$u_{tt} = \Delta u, 0 < r < 1, u|_{r=1} = 0, \ u|_{t=0} = r, u_t|_{t=0} = 1. \tag{8}$$

Proof (Hernandez). Our equation is $u_{tt} = \Delta u = u_{rr} + \frac{2}{r}u_r$, giving a solution of the following form:

$$u(r,t) = \sum_{n=1}^{\infty} c_n(t) y_n(r), \qquad y_n(r) = \frac{\sin(n\pi r)}{r},$$

and eventually obtain:

$$u(r,t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2T_0}{n} e^{-n^2 kt} \frac{\sin(nr)}{r}.$$

□